GRIET — Department of Computer Science and Engineering
Mini Project 2024–25 · Batch A9

COMORA

Where People Meet Around Ideas

Software Requirements Specification (SRS)

Version 1.0 · March 2025

Document Type	Software Requirements Specification
Project	Comora — Agenda-First Social Gathering Platform
Version	1.0
Status	Draft — Active Development
Platform	Web Application (Vibe Coded · Free-Tier Stack)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for Comora — an agenda-first, interest-driven social gathering platform. The document is intended to serve as the authoritative technical and functional reference throughout the entire development lifecycle, from initial vibe-coded prototyping through phased feature delivery.

This document must be read in conjunction with the Comora Concept & Vision Document (2024–25), which establishes the philosophical foundation and strategic direction of the platform. Where the Concept Document defines the why, this SRS defines the what and how — translated into implementable, testable requirements.

Primary audiences for this SRS include the development team, the project guide (V. Lakshmi, AP), and any future contributors or evaluators reviewing the project.

1.2 Project Scope

Comora is a web-based platform that enables individuals to create, discover, and attend purpose-driven social gatherings centered around shared intellectual and social interests — not food or dining. The platform serves three primary user roles: Guests (attendees), Hosts (event curators), and Admins (platform operators).

The platform provides:

- A structured event creation and management system for Hosts
- An interest-based discovery and booking system for Guests
- An AI-assisted content and recommendation layer
- A trust and safety infrastructure including verification, escrow, and ratings
- An administrative panel for platform moderation, analytics, and user management

The following are explicitly OUT OF SCOPE for this version:

- Native mobile applications (iOS / Android)
- Real-money payment processing (mocked/simulated in free-tier MVP)
- Live video or streaming capabilities
- Multi-language support (English only at launch)

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
Comora	The platform — an agenda-first social gathering product for Indian urban communities
SRS	Software Requirements Specification — this document
Guest	A registered user who discovers and attends events created by Hosts

Term / Acronym	Definition
Host	A registered user who creates, curates, and manages Comora events
Admin	A platform-level operator with privileged access for moderation and oversight
Event	A structured social gathering listed on the platform with an agenda, time, and guest capacity
Agenda	The stated intellectual or social purpose of an event (e.g., book discussion, networking, film review)
Agenda Card	The visual summary block of an event's purpose, vibe, format, and audience type
Agenda Pack	A reusable, multi-session event template created and optionally published by a Host
Community	A recurring group of Hosts and Guests centred around a shared topic or interest
Match Me	The onboarding quiz that builds a guest's interest and social profile
Host Studio	The guided event-creation tool available to Hosts
AI Assist	Gemini API-powered features embedded within the platform
RSVP	Reservation system through which Guests confirm attendance at an event
Escrow	Simulated payment hold mechanism — funds held until event completion
MVP	Minimum Viable Product — the initial deployable version of the platform
Free-Tier	All infrastructure and third-party services used at zero cost (Vercel, Supabase, etc.)
Vibe Coding	AI-assisted rapid prototyping approach used for development via Claude AI (for coding) and Gemini API (for in-app AI features)
FR	Functional Requirement
NFR	Non-Functional Requirement
UI	User Interface
UX	User Experience
API	Application Programming Interface
JWT	JSON Web Token — used for authentication sessions
LLM	Large Language Model — used for AI Assist features

1.4 Document Overview

This SRS is organized into seven major sections, each of which corresponds to a distinct dimension of the system:

1. Introduction — Document purpose, scope, terminology, and structure
2. Overall System Description — Product context, user roles, personas, and system constraints
3. System Features — Core functional requirements for Guest, Host, Admin, and AI features
4. External Interface Requirements — UI, API, hardware, and communication specifications
5. Non-Functional Requirements — Performance, security, scalability, and quality constraints
6. Technology Stack & System Constraints — Tools, frameworks, and free-tier limitations
7. Project Roadmap & Delivery Phases — Phased development and milestone plan

2. Overall System Description

2.1 Product Perspective

Comora is an independently developed web application with no direct dependency on or integration with any existing large-scale platform. It is designed as a standalone social infrastructure product, similar in category to Eventbrite or Meetup, but differentiated by its core agenda-first philosophy and AI-assisted community curation layer.

The system comprises three interconnected modules that together form the complete platform:

> *Guest Module ↔ Host Module ↔ Admin ModuleAll three modules share a common backend (Supabase) and authentication layer, with role-based access controls governing data visibility and action permissions.*

The AI Assist layer (powered by Google Gemini API — free tier) is a cross-cutting capability that surfaces within both the Guest and Host modules to enhance content quality, reduce friction, and personalise the user experience.

2.2 Product Functions Summary

Function Area	Guest Experience	Host Experience
Discovery	Browse, search, and filter events by topic, vibe, format, and location	Manage event visibility; view who is viewing your listing
Onboarding	Match Me profile quiz for personalised recommendations	Host profile setup with verification steps
Events	View agenda cards, RSVP, receive pre-event packs	Create events via Host Studio; manage registrations
AI Features	Explainable recommendations, icebreaker packs	AI theme-to-menu generator, screening question builder
Community	Join topic-based communities, follow hosts, view event series	Build recurring series, publish Agenda Packs
Safety	Guest verification, pre-event briefings	Host verification badge, escrow-based payments
Post-Event	Rate host + event on multi-axis; connect with co-attendees	Receive structured feedback; build reputation score
Admin	Platform-wide moderation, analytics, content review	Host approval workflows, dispute resolution

2.3 User Roles and Personas

2.3.1 Guest (Attendee)

Guests are the primary consumers of Comora events. They discover, book, and attend gatherings created by Hosts.

Attribute	Description
Primary Goal	Find social gatherings that match their specific intellectual or social interests
Pain Points	Unable to find structured, interest-driven events; general social anxiety at unstructured events
Demographics	College students, young urban professionals (20–32), new-to-city individuals, introverts
Tech Literacy	Moderate to high; comfortable with web apps and digital payments
Key Motivators	Genuine connection, intellectual stimulation, community belonging, reduced awkwardness
Persona Example	Priya, 24, new to Hyderabad for work. Loves fantasy fiction. Can't find her tribe.

2.3.2 Host (Event Curator)

Hosts are platform creators who design and facilitate social gatherings around their areas of passion and expertise.

Attribute	Description
Primary Goal	Build a recurring community around a topic they care deeply about
Pain Points	Logistics of managing events are complex; no dedicated platform for agenda-first gatherings
Demographics	Enthusiasts, domain experts, freelancers, educators, artists, community organisers (22–40)
Tech Literacy	Moderate; may not be highly technical but comfortable with forms, dashboards, and messaging tools
Key Motivators	Community recognition, supplemental income, sharing expertise, legacy building
Persona Example	Arjun, 29, literature lover and part-time copywriter. Runs a monthly fiction circle from his apartment.

2.3.3 Admin (Platform Operator)

Admins are the internal team managing the health, safety, and growth of the Comora platform.

Attribute	Description
Primary Goal	Maintain platform trust, safety, and quality; monitor growth metrics
Responsibilities	Host verification, content moderation, dispute resolution, analytics review
Access Level	Full access to all user data, event content, financial records, and system settings
Persona Example	Veeraj / core team members managing the platform during MVP phase

2.3.4 Unregistered Visitor (Guest)

Users who land on Comora without an account. They can browse public event listings and landing pages but cannot book, RSVP, or access community features. Their experience is designed to drive account creation.

2.4 Operating Environment

- Web-first: optimised for desktop and mobile browsers (Chrome, Firefox, Safari)
- No native app for MVP; responsive design handles mobile use cases
- Hosted on Vercel (free tier) with Supabase as BaaS (Backend as a Service)
- Google Gemini API (free tier via Google AI Studio) used for all AI Assist features — no credit card required, generous daily quota sufficient for MVP
- All third-party services used must have a functional free tier (no credit card required for MVP)

2.5 Design and Implementation Constraints

- All infrastructure must remain within free-tier service limits throughout the college project lifecycle
- Development methodology: Vibe Coding using Claude AI — rapid iteration, AI-generated boilerplate, human review and customisation
- No native payment gateway integration in MVP; payment is simulated via a form-based escrow placeholder
- No server-side rendering frameworks required; Next.js or React SPA acceptable
- Team of 4 with mixed technical background — complexity must remain manageable within a semester
- All data stored in Supabase PostgreSQL with Row-Level Security (RLS) enforcing role-based access

3. System Features & Functional Requirements

Feature IDs follow the format [ROLE]-[CATEGORY]-[NUMBER]. Priority levels: P1 = MVP / Must Have | P2 = Phase 2 / Should Have | P3 = Future / Nice to Have

3.1 Guest Features

3.1.1 Authentication & Onboarding

The onboarding experience is Comora's most important first impression. It must feel fast, conversational, and rewarding — not like filling out a form.

Feature ID	Feature Name	Description	Priority
G-AUTH-01	Registration	Email-based account creation with name, profile photo, and city	P1
G-AUTH-02	Login / Logout	Session-based login via JWT; persistent session across tabs	P1
G-AUTH-03	Match Me Onboarding	60–90 sec guided profile quiz capturing interests, social comfort, group size preference, food prefs, budget, and location radius. Completed at account creation.	P1
G-AUTH-04	Profile Management	Edit name, photo, bio, interest tags, social comfort level, dietary preferences, and city	P1
G-AUTH-05	Google OAuth (SSO)	Sign in with Google for frictionless onboarding	P2
G-AUTH-06	Guest Verification	Optional verification tiers: Light (email confirmed), Medium (phone + photo), Strong (ID upload for private events)	P2

3.1.2 Event Discovery

Discovery is the core value loop for Guests. It must be interest-led, not cuisine-led, and must surface events that feel personally curated.

Feature ID	Feature Name	Description	Priority
G-DISC-01	Browse Events	Homepage feed of upcoming events; default sorted by recommendation score	P1
G-DISC-02	Agenda-First Filters	Filter events by topic/fandom, vibe (quiet/loud, structured/casual), format (discussion/workshop/screening), group size, date, and city	P1
G-DISC-03	Search	Full-text search across event titles, agenda descriptions, and topic tags	P1

Feature ID	Feature Name	Description	Priority
G-DISC-04	Agenda Card View	Each event displays: What we'll do, Vibe level, Who it's for, Format type, Group size, Price, Host name + rating	P1
G-DISC-05	AI Recommendations	Personalised event feed based on Match Me profile with explainable reasoning labels ('Because you like X and prefer Y')	P2
G-DISC-06	Topic Communities	Browse by community/circle (e.g., 'LOTR Circle – Hyderabad', 'Indie Music – Bengaluru')	P2
G-DISC-07	Follow Host	Follow a Host to get notified about their upcoming events	P2
G-DISC-08	Event Series Browser	View and follow multi-session agenda packs (e.g., 'LOTR Night v1, v2, v3')	P3

3.1.3 Event Booking & RSVP

Feature ID	Feature Name	Description	Priority
G-BOOK-01	View Event Details	Full event page: agenda, host profile, venue details, attendee count, dietary info, reviews	P1
G-BOOK-02	RSVP / Book	One-click RSVP for free events; simulated payment confirmation form for paid events	P1
G-BOOK-03	Waitlist	Auto-join waitlist when event is full; notification on seat opening	P1
G-BOOK-04	Cancellation	Guest can cancel RSVP; cancellation policy displayed at booking (e.g., 48-hour refund window)	P1
G-BOOK-05	My Bookings Dashboard	View upcoming and past bookings with status badges (Confirmed / Waitlisted / Cancelled / Attended)	P1
G-BOOK-06	Event Reminders	Automated email reminders at T-48h and T-2h before event	P2
G-BOOK-07	Bring-a-Friend Invite	Invite a friend to an event only if their profile matches the event's interest tags	P3

3.1.4 Pre-Event & Social Tooling

Feature ID	Feature Name	Description	Priority
G-SOC-01	Conversation Warm-Up Pack	Unlocked after booking: AI-generated icebreaker questions, discussion prompts, and a mini trivia round relevant to the event's agenda	P1 (AI)

Feature ID	Feature Name	Description	Priority
G-SOC-02	Seat Mixing Preference	Guests choose: 'Meet new people' or 'Sit near similar profiles'	P2
G-SOC-03	Attendee Preview	See anonymised attendee tags (not names) before the event — e.g., '3 designers, 2 booklovers'	P2
G-SOC-04	Post-Event Connection	After attending, guests can choose to connect with co-attendees on-platform before optionally exchanging external contact details	P2
G-SOC-05	Season Pass	Subscribe to a topic cluster (e.g., '3 events/month in Fantasy Fiction')	P3

3.1.5 Post-Event & Reputation

Feature ID	Feature Name	Description	Priority
G-REV-01	Multi-Axis Rating	After attending: rate Agenda Quality, Host Warmth, Food Accuracy, and Group Vibe on 5-point scales with optional text comment	P1
G-REV-02	RSVP Reliability Score	System tracks guest cancellations and no-shows; score displayed to Hosts during registration screening	P2
G-REV-03	Post-Event Memory	Opt-in auto-generated event summary: key discussion takeaways + shared content from host (photo album, recipe card)	P3

3.2 Host Features

3.2.1 Host Registration & Verification

Feature ID	Feature Name	Description	Priority
H-AUTH-01	Host Registration	Upgrade from Guest account or register directly as Host; requires name, phone, city, bio, and area of expertise	P1
H-AUTH-02	Host Verification	Mandatory: submit government ID and phone number. Platform displays a Verified Host badge after admin review	P1
H-AUTH-03	Host Profile	Public-facing profile: photo, bio, topic tags, past event history, average ratings, and upcoming events	P1
H-AUTH-04	Venue Declaration	Host submits venue address/type (home/café/community hall) visible to confirmed guests only	P1

3.2.2 Host Studio — Event Creation

The Host Studio is the core creation tool. It is a guided, step-by-step builder that reduces friction and produces high-quality, consistent event listings.

Feature ID	Feature Name	Description	Priority
H-STUDIO-01	Agenda Template Picker	Choose from templates: Book Club, Debate, Workshop, Film Screening, Storytelling, Networking, Tasting, Open Discussion	P1
H-STUDIO-02	Vibe Sliders	Set event parameters: Structured↔Freeform, Quiet↔Loud, Beginner↔Expert, Small↔Large group	P1
H-STUDIO-03	Food Layer Setup	Optional: add cuisine type, dietary accommodations, and let AI suggest a themed menu to match the agenda	P1
H-STUDIO-04	Event Details Form	Title, description, date/time, duration, venue type, max guest count, price per seat, cancellation policy	P1
H-STUDIO-05	Registration Mode	Choose: Open (anyone can book), Request-to-Join (host approves), Invite-Only (host sends invitations)	P1
H-STUDIO-06	AI Description Enhancer	Paste a rough event description; AI refines it for tone, clarity, and safety alignment	P1 (AI)
H-STUDIO-07	Soft Screening Questions	AI generates 2–3 interest-based questions for Request-to-Join events (e.g., 'What's your favourite Tolkien arc?')	P2 (AI)
H-STUDIO-08	Preview & Publish	Preview the Agenda Card as guests will see it; confirm and publish or save as draft	P1
H-STUDIO-09	Edit / Cancel Event	Edit event details before a booking threshold; cancel with automated guest notification	P1

3.2.3 AI Theme-to-Menu Generator

This is Comora's signature AI feature — the one that feels 'magical' because it reduces real, meaningful work for the host.

Feature ID	Feature Name	Description	Priority
H-AI-01	Theme Input Prompt	Host enters a natural language prompt: e.g., 'LOTR book lovers, cozy winter vibe, vegetarian friendly'	P1 (AI)
H-AI-02	Themed Menu Generation	AI returns: course names with thematic names (e.g., 'The Shire's Harvest Soup'), ingredient list, dietary swaps	P1 (AI)

Feature ID	Feature Name	Description	Priority
H-AI-03	Prep Timeline	AI generates estimated prep timeline and shopping list based on the generated menu	P2 (AI)
H-AI-04	Story Moments	AI suggests optional mini-activities or discussion triggers to place between courses	P2 (AI)
H-AI-05	Agenda Pack Drafts	AI assists Host in drafting a multi-session series (v1, v2, v3...) with thematic continuity	P3 (AI)

3.2.4 Host Dashboard & Operations

Feature ID	Feature Name	Description	Priority
H-OPS-01	Guest List Management	View confirmed/waitlisted/cancelled guest list with RSVP reliability scores and dietary notes	P1
H-OPS-02	Request-to-Join Approvals	Review guest applications for gated events; approve/decline with optional personalised note	P1
H-OPS-03	Automated Reminders	System sends WhatsApp/email reminders to confirmed guests (T-48h, T-2h); host can add a custom message	P2
H-OPS-04	Smart Event Checklist	Auto-generated pre-event checklist: cookware, seating plan, dietary flags, agenda run-of-show timer	P2
H-OPS-05	Run-of-Show Timer	In-event tool: countdown timer for agenda segments (e.g., 'Welcome 10m → Discussion 45m → Debrief 15m')	P2
H-OPS-06	Earnings Overview	Summary of confirmed bookings, simulated revenue, and platform fee deduction	P1

3.2.5 Host Reputation & Analytics

Feature ID	Feature Name	Description	Priority
H-REP-01	Multi-Axis Rating Receipt	View detailed ratings from each event: Agenda Quality, Warmth, Food Accuracy, Group Vibe, + text comments	P1
H-REP-02	Pricing Intelligence	Show estimated fill rate at different price points based on comparable events on the platform	P2
H-REP-03	Group Composition Advisor	AI suggests an ideal guest mix based on event type (introvert/extrovert balance, new/returning attendees)	P2
H-REP-04	Event Analytics	Views, RSVP conversion rate, repeat guest rate, cancellation rate per event	P2

3.2.6 Agenda Packs (Series Management)

Feature ID	Feature Name	Description	Priority
H-PACK-01	Create Agenda Pack	Group related events into a named series (e.g., 'LOTR Night: Fellowship → Two Towers → Return')	P2
H-PACK-02	Publish Pack	Make the Agenda Pack publicly discoverable; guests can follow the series and get auto-notified of new sessions	P3
H-PACK-03	Pack Analytics	Track series follow-through rate: what % of guests who attended v1 came back for v2	P3

3.3 Admin Features

3.3.1 User & Host Management

Feature ID	Feature Name	Description	Priority
A-USER-01	User Directory	Search, filter, and view all registered users with role, status, and activity summary	P1
A-USER-02	Host Verification Queue	Review pending Host applications; approve/reject with a reason; manage Verified badges	P1
A-USER-03	Suspend / Ban User	Temporarily suspend or permanently ban a user from the platform with an audit log entry	P1
A-USER-04	Role Management	Promote a Guest to Host; demote a Host; assign Admin roles	P1

3.3.2 Event Moderation

Feature ID	Feature Name	Description	Priority
A-MOD-01	Event Review Queue	All new events require admin approval before going live (configurable to auto-approve after host trust threshold)	P1
A-MOD-02	Content Flagging	Review user-reported events, hosts, or reviews; resolve or escalate	P1
A-MOD-03	Event Takedown	Force-cancel a live event with guest notification and simulated refund trigger	P1
A-MOD-04	AI Content Scan	AI auto-flags event descriptions containing prohibited content (violence, misinformation, explicit content) before publish	P2 (AI)

3.3.3 Platform Analytics Dashboard

Feature ID	Feature Name	Description	Priority
A-ANALYTICS-01	User Growth	DAU/MAU, new registrations per week, Guest-to-Host conversion rate	P1
A-ANALYTICS-02	Event Metrics	Events published, events completed, average fill rate, cancellation rate	P1
A-ANALYTICS-03	Revenue Simulation	Total simulated platform revenue, top-earning hosts, average event price	P2
A-ANALYTICS-04	Community Health	Community size, average recurring attendance rate, most active topics	P2
A-ANALYTICS-05	Geographic Heatmap	City-level event density and user concentration map	P3

3.3.4 Trust & Safety Tools

Feature ID	Feature Name	Description	Priority
A-TRUST-01	Dispute Resolution Console	View dispute submissions from Guests or Hosts; record resolution and trigger escrow action	P1
A-TRUST-02	Escrow Management	Simulated escrow overview: held funds, release triggers, refund log	P1
A-TRUST-03	Platform Code of Conduct	Display, version-control, and enforce platform terms; track user acknowledgement	P1
A-TRUST-04	Audit Log	Immutable log of all admin actions (user bans, event removals, escrow releases) with timestamps	P2

3.4 AI Feature Summary (Google Gemini API Integration)

Comora integrates the Google Gemini API (free tier, via Google AI Studio) as its AI backbone. All AI features are designed to be clearly useful, contextually appropriate, and technically feasible within free-tier API limits. Each AI feature reduces genuine friction for either the Guest or the Host.

Feature ID	Feature	User Role & Trigger	Priority
AI-01	Match Me Profile Analysis	Guest — on onboarding completion; builds interest vector for recommendations	P1
AI-02	Explainable Recommendations	Guest — on homepage; labels why each event is recommended ('Because you like X')	P1
AI-03	Warm-Up Icebreaker Pack	Guest — after RSVP confirmed; Gemini generates 5 icebreakers + 3 discussion prompts relevant to the agenda	P1
AI-04	Theme-to-Menu Generator	Host — in Host Studio; Host inputs theme + constraints; Gemini outputs menu names, shopping list, dietary swaps	P1
AI-05	Event Description Enhancer	Host — in Host Studio; Host pastes a rough description; Gemini refines tone, clarity, and safety	P1
AI-06	Screening Question Generator	Host — for Request-to-Join events; Gemini generates interest-based screening questions from the agenda tags	P2
AI-07	Group Composition Advisor	Host — in dashboard; Gemini suggests ideal new/returning guest mix based on event format	P2
AI-08	Post-Event Takeaway Summary	Guest (opt-in) — after event; Gemini generates key discussion takeaways + memory card	P2
AI-09	Admin Content Safety Scan	Admin — on event publish request; Gemini scans description for policy violations	P2
AI-10	Pricing Intelligence Insights	Host — in dashboard; Gemini interprets comparable event data to suggest optimal pricing	P3

Implementation note: All Gemini API calls use a defined system prompt to ensure consistent output format and prevent hallucinations. Streaming responses are used where UX benefits from progressive text display (e.g., menu generation). Max tokens set conservatively to stay within free-tier daily quota (1,500 requests/day on Gemini 2.5 Flash free tier).

4. External Interface Requirements

4.1 User Interface Requirements

- The application must be fully responsive and usable on screens from 375px (mobile) to 1440px (desktop) without horizontal scrolling
- The design system must be consistent with Comora's brand: clean editorial aesthetic, warm navy + steel blue palette, high whitespace
- All primary actions (RSVP, Publish, Submit) must be reachable within 2–3 taps/clicks from the relevant feature entry point
- Loading states must be indicated for all async operations (skeleton loaders or spinners)
- Error messages must be human-readable, contextual, and actionable — not raw API error codes
- Forms must include inline validation with clear field-level error states
- The Match Me onboarding flow must feel conversational (one question per screen, progress bar visible)
- Agenda Card design must prioritise topic, vibe, and format over food/price — food details appear below the fold

4.2 API Interfaces

API / Service	Usage in Comora	Integration Method
Google Gemini API	All AI Assist features: recommendations, menu gen, description enhancer, icebreakers, screening questions, content safety scan	REST via fetch(); API key in server-side env variable (Supabase Edge Function); Free tier via Google AI Studio — no credit card required
Supabase REST API	All CRUD operations: users, events, bookings, reviews, communities	Supabase JS Client SDK
Supabase Auth	JWT-based auth, session management, email verification, OAuth hooks	Supabase Auth SDK
Supabase Storage	Host/guest profile photos, event images, uploaded verification documents	Supabase Storage SDK
Supabase Realtime	Live updates on event capacity, waitlist status, and admin alerts	Supabase Realtime subscriptions
Resend / Nodemailer	Transactional emails: booking confirmation, reminders (T-48h, T-2h), cancellation	REST API / SMTP (free tier)
Google Maps Embed API	Static map embed on event detail page showing approximate venue area	Iframe embed (free quota)

4.3 Hardware Interfaces

- No dedicated hardware interface requirements — standard web browser on any device
- Camera/photo access required only for profile photo upload (browser FileAPI)
- GPS/Location — browser Geolocation API used optionally to auto-fill city on registration

4.4 Communication Interfaces

- HTTPS mandatory for all client-server communication; HTTP connections automatically redirected
- WebSocket connections used for Supabase Realtime features (waitlist, capacity updates)
- Email notifications via SMTP relay (Resend free tier: 100 emails/day, 3,000/month)
- No in-app messaging or chat in MVP; community interaction routed through post-event connections only

5. Non-Functional Requirements

5.1 Performance

Requirement	Target / Constraint
Initial Page Load (LCP)	< 2.5 seconds on 4G mobile connection
API Response Time	< 500ms for standard database queries; < 3s for Gemini API calls
Event Listing Load	Homepage event feed renders within 1.5 seconds (paginated, 10 events per page)
Image Optimisation	Profile and event images compressed and served via Supabase CDN in WebP format
Concurrent Users (MVP)	Designed to support up to 500 concurrent users within Supabase free-tier limits

5.2 Security

- All API keys (Gemini, Supabase service role) stored exclusively in server-side environment variables — never exposed to client
- Row-Level Security (RLS) enabled on all Supabase tables; Guests cannot read other users' private data
- JWT tokens expire after 24 hours; refresh token rotation enabled
- Input sanitisation on all user-generated content fields to prevent XSS and SQL injection
- File uploads restricted to image types (PNG, JPG, WEBP); max 5MB; scanned for MIME type mismatch
- Admin panel routes protected by both role check (Supabase RLS) and server-side middleware
- Password hashing via bcrypt (handled by Supabase Auth)
- Venue addresses encrypted at rest; only revealed to confirmed guests

5.3 Scalability

- Stateless application architecture enables horizontal scaling without session stickiness issues
- Database schema designed with indexing on frequently filtered fields: city, topic_tags, event_date, status
- Gemini API calls are async and non-blocking; failures gracefully degrade to manual input mode
- Supabase free tier supports up to 500MB database and 1GB file storage — sufficient for MVP phase
- Pagination implemented on all list views (events, guests, reviews) to prevent memory overload

5.4 Reliability & Availability

- Platform targets 99.5% uptime leveraging Vercel's global CDN and Supabase's managed infrastructure
- All form submissions include optimistic UI with server-side confirmation to prevent duplicate submissions
- Supabase database includes automated daily backups (free tier: 7-day retention)
- If Gemini API is unavailable or daily quota is reached, AI Assist features display a graceful fallback message; no critical path blocked

5.5 Usability

- WCAG 2.1 Level AA accessibility compliance for all primary user flows (colour contrast, keyboard navigation, ARIA labels)
- Onboarding completion rate target: > 70% of users who start Match Me should complete all steps
- First-event booking target: > 40% of new registered Guests should complete a booking within their first session
- Error recovery: all form errors should be recoverable without page refresh or data loss
- All primary features must be usable without reading any documentation or help text

5.6 Maintainability

- Codebase structured as modular React components with clear separation of concerns
- All Gemini API prompts stored as named constants in a dedicated prompts.js file for easy iteration
- Database schema changes managed via Supabase migrations with version control
- README documentation to cover local setup, environment variable configuration, and deployment steps
- All AI-generated code reviewed and commented by the team before merge

6. Technology Stack & System Constraints

All services listed below must remain within free-tier limits for the duration of the college project. No credit card billing should be triggered.

6.1 Frontend

Technology	Role	Free Tier Notes
React.js (Vite)	Core UI framework — component-based SPA	Open source — no cost
Tailwind CSS	Utility-first styling — fast, consistent design system	Open source — no cost
React Router	Client-side routing between pages	Open source — no cost
React Query / SWR	Server state management, caching, and background refetching	Open source — no cost
Framer Motion	Micro-animations: page transitions, skeleton loaders	Open source — no cost
Lucide React	Icon library — consistent visual language	Open source — no cost

6.2 Backend / BaaS

Technology	Role	Free Tier Notes
Supabase (Free Tier)	PostgreSQL database, Auth, Storage, Realtime, Edge Functions	500MB DB, 1GB Storage, 50,000 MAU — sufficient for MVP
Supabase Auth	JWT authentication, email/password, OAuth (Google)	Included in free tier
Supabase Storage	Profile photos, event images, verification documents	1GB included
Supabase Edge Functions	Server-side logic for Gemini API calls (keeps key secure)	500,000 invocations/month free
Supabase Realtime	Live updates on event capacity and waitlist	200 concurrent connections free

6.3 AI Layer

Comora uses the Google Gemini API (free tier via Google AI Studio) as its sole AI provider. No credit card is required. The primary model is Gemini 2.5 Flash. Groq is included as a fallback provider for latency-sensitive features. All API calls are routed through Supabase Edge Functions so keys remain server-side and never exposed to the browser.

Technology	Role in Comora	Free Tier Notes
Google Gemini API (Primary)	All 10 AI Assist features: menu generation, icebreakers, description enhancer, recommendations, screening questions, content safety scan	Free via Google AI Studio. No credit card. 1,500 requests/day on Gemini 2.5 Flash free tier
Gemini 2.5 Flash (Model)	Recommended model — best balance of quality, speed, and free-tier quota for all Comora use cases	Free. Multimodal, fast inference, supports JSON-mode output for structured AI responses
Groq API (Fallback)	Fallback for latency-sensitive features (e.g., real-time icebreaker generation) when Gemini quota is reached. Runs Llama 3.3 70B	Free tier. No credit card. 300+ tokens/second inference speed. OpenAI-compatible API format
prompts.js (In-codebase)	All system prompts stored as named constants. Controls output format, prevents hallucinations, easy to iterate during vibe coding	No cost — in-codebase file. Version controlled via GitHub
Supabase Edge Functions (Proxy)	Secure server-side proxy for all Gemini and Groq API calls. GEMINI_API_KEY and GROQ_API_KEY stored as Supabase secrets — never in frontend code	500,000 invocations/month free on Supabase free tier

6.4 Deployment & DevOps

Technology	Role	Free Tier Notes
Vercel (Hobby Tier)	Frontend hosting with global CDN; CI/CD from GitHub	Unlimited hobby deployments; 100GB bandwidth/month free
GitHub	Version control, collaboration, project board	Free for teams up to 3; unlimited public/private repos
GitHub Actions	Automated lint + build checks on every PR	2,000 CI minutes/month free
Vercel Analytics	Page view and performance monitoring	Included in Hobby tier

6.5 Communication & Notifications

Technology	Role	Free Tier Notes
Resend	Transactional email: booking confirmations, reminders	3,000 emails/month; 100/day — sufficient for MVP
Google Maps Embed API	Static venue area map on event detail page	Free quota: 28,000 map loads/month

6.6 Database Schema Overview

The following tables form the core relational schema in Supabase PostgreSQL:

Table Name	Key Fields / Purpose
users	id, name, email, role (guest/host/admin), city, bio, interests[], social_comfort, dietary_prefs, verification_level, created_at
events	id, host_id, title, description, agenda_type, vibe_tags[], topic_tags[], format, date_time, duration, venue_type, venue_address_encrypted, max_guests, price, status, registration_mode
bookings	id, event_id, guest_id, status (confirmed/waitlisted/cancelled), payment_status (simulated), booked_at
communities	id, name, topic_tags[], city, created_by, member_count, created_at
community_members	id, community_id, user_id, joined_at
reviews	id, event_id, reviewer_id, agenda_quality, host_warmth, food_accuracy, group_vibe, comment, created_at
agenda_packs	id, host_id, series_name, description, session_count, is_published
notifications	id, user_id, type, message, is_read, created_at
disputes	id, event_id, raised_by, against, description, status, resolved_by, created_at
audit_logs	id, admin_id, action_type, target_type, target_id, details, created_at

7. Project Roadmap & Delivery Phases

The Comora development roadmap is structured into four phases, aligned with a college semester timeline and the free-tier infrastructure constraints. Each phase delivers a shippable increment.

Phase 1 — Foundation & Core MVP (Weeks 1–3)

Goal: A working platform where a Host can create an event and a Guest can discover and RSVP to it.

Deliverable	Details
Project Setup	GitHub repo, Vercel deployment, Supabase project, environment variables configured
Database Schema	Core tables: users, events, bookings created with RLS policies
Authentication	Email/password registration and login via Supabase Auth; JWT session management
Match Me Onboarding	Guest profile quiz — 6 steps, interests and preferences captured and stored
Host Registration	Host signup with profile fields; admin verification queue placeholder
Basic Event Creation	Host Studio v1 — title, description, date, price, max guests, topic tags
Event Listing Page	Homepage with event cards; topic filter and search bar
Agenda Card	Event detail page with agenda, vibe indicators, and host summary
RSVP / Book Flow	One-click RSVP for free events; simulated payment form for paid events
My Bookings Dashboard	Guest view of upcoming/past bookings with status badges

Phase 2 — AI Features & Host Studio (Weeks 4–6)

Goal: Make the platform feel intelligent. Integrate Google Gemini API for the core AI features on both sides.

Deliverable	Details
Gemini API Integration	Supabase Edge Function as a secure proxy for all Gemini API calls

Deliverable	Details
AI Description Enhancer	Host Studio: rough-to-polished event description generation (H-STUDIO-06)
Theme-to-Menu Generator	Host AI feature: theme prompt → menu names + shopping list (H-AI-01 to H-AI-02)
AI Warm-Up Icebreaker Pack	Guest receives icebreakers + discussion prompts post-RSVP (G-SOC-01)
Explainable Recommendations	Homepage AI feed with 'Because you like X' labels (G-DISC-05)
Vibe Sliders in Host Studio	Structured↔Freeform, Quiet↔Loud, Beginner↔Expert sliders (H-STUDIO-02)
Registration Mode	Open / Request-to-Join / Invite-Only selection (H-STUDIO-05)
Multi-Axis Rating System	Post-event review form (4 axes + text); ratings visible on host profile (G-REV-01)
Host Verification Queue	Admin can approve/reject host verification applications (A-USER-02)

Phase 3 — Community, Safety & Admin Panel (Weeks 7–9)

Goal: Build trust infrastructure, community features, and operational admin tools.

Deliverable	Details
Admin Dashboard	User directory, event moderation queue, dispute console, platform stats (A-USER, A-MOD)
Host Analytics	Event-level analytics: views, RSVP rate, repeat guests, cancellation rate (H-REP-04)
Topic Communities	Community pages with member lists and event feeds (G-DISC-06)
Waitlist System	Auto-waitlist on full events with seat-opening notifications (G-BOOK-03)
Event Reminder Emails	T-48h and T-2h reminders via Resend integration (G-BOOK-06, H-OPS-03)
RSVP Reliability Score	System tracks guest cancellations; score visible to hosts (G-REV-02)
Simulated Escrow	Payment hold and release logic with dispute trigger (A-TRUST-02)
AI Screening Questions	AI generates interest-based screening Qs for Request-to-Join events (H-STUDIO-07)

Deliverable	Details
Post-Event Connections	Guests can choose to connect on-platform with co-attendees after event (G-SOC-04)

Phase 4 — Polish, Testing & Future Scope (Weeks 10–12)

Goal: Harden the platform, fix bugs, improve UX, and prepare documentation + demo for evaluation.

Deliverable	Details
Full Responsive QA	Test all flows on mobile (375px), tablet (768px), and desktop (1440px)
Accessibility Audit	WCAG 2.1 AA compliance check on all primary user flows
Performance Optimisation	Image compression, lazy loading, pagination audit, Lighthouse score target ≥ 85
Agenda Packs (Series)	Host can create and publish multi-session series (H-PACK-01, H-PACK-02) — P3
Season Pass	Guest subscribes to 3-events/month topic cluster (G-SOC-05) — P3
Google OAuth	Sign in with Google for frictionless onboarding (G-AUTH-05) — P2
Post-Event Memory Card	Opt-in AI-generated event summary + takeaways (G-REV-03, AI-08) — P2/P3
Demo Preparation	Clean seed data, walkthrough script, project report, and presentation deck
Concept + SRS Alignment Check	Verify all Concept Document pillars are reflected in the live product

7.1 Feature Priority Summary

The table below provides a consolidated view of all features by priority tier:

Priority	Label	Scope
P1 — Must Have	MVP — Phases 1 & 2	Core registration, event CRUD, RSVP, AI description enhancer, theme-to-menu generator, icebreaker pack, basic admin
P2 — Should Have	Phase 3	AI recommendations, communities, waitlist, reminders, analytics,

Priority	Label	Scope
		screening questions, escrow, post-event connections
P3 — Nice to Have	Phase 4 / Post-MVP	Agenda Packs, Season Passes, Google OAuth, Memory Cards, Geographic Heatmap, AI pricing intelligence

This SRS is a living document. It should be updated as the team iterates on features during vibe-coding sessions with Claude AI. Any deviation from a P1 requirement must be documented as a known limitation in the project report.

— End of Document —